Stochastic Calculus and Option Pricing
Lecture 12 Exercises

12 Week 12 — Beyond Black–Scholes

Focus. The questions isolate covariance in random discounting and the risk-spanning consequence of adding a volatility factor.

Exercise 12.1 (Random discounting and conditional covariance)**.** Under a risk-neutral measure, let

$$D_{t,T} = \exp\left(-\int_t^T r_u \, \mathrm{d}u\right)$$

and let a claim pay $H_T \geq 0$ at T.

(a) Write its time-t value as a conditional expectation and decompose it into a product of conditional means plus a conditional covariance term.

(b) Give a sufficient condition under which replacing the random discount factor by its conditional mean times the expected payoff is valid.

(c) Explain why a positive covariance between discounting and payoff raises the value relative to that product approximation.

Exercise 12.2 (A second Brownian shock adds a hedge direction)**.** Suppose $V_t > 0$, and the stock and variance satisfy

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sqrt{V_t}\,\mathrm{d}W_t^S, \qquad \mathrm{d}V_t = b(t,V_t)\,\mathrm{d}t + \eta(t,V_t)\,\mathrm{d}W_t^V,$$

with $|\rho| < 1$ between W^S and W^V, and $\eta \neq 0$.

(a) Explain why the two Brownian shocks span two independent local risk directions.

(b) A desk trades only the stock and cash. Which risk direction can it hedge instantaneously, and which remains?

(c) What feature would a second traded instrument need in order to complete the local hedge? State the condition in terms of its exposure vector, rather than naming a particular product.